

## Advanced Econometrics - Lecture 10

### Exercise 2 from Advanced Econometrics Exam (2014)

Lecture given in Polish. Copyright©2014 dr Jerzy Mycielski.

Using data from the Polish General Social Survey (PGSS) from 1999, 2002, 2005, 2008, and 2010 an economist tried to identify variables that affect education level of respondents. The dependent variable in the model is *educ*, which takes: 1 for people with primary education only, 2 for secondary education only, and 3 for higher education. As covariates the economist decided to use: *female* (1 – female, 0 – male), *age*, *age2* ( $age^2$ ), and *pgssyear* denoting a year of survey. Below, you may find either parameters' estimators or marginal effects for the alternative *higher education* that both come from ordered probit model.

| educ                   | parameter | marginal<br>effect | z     | p     |
|------------------------|-----------|--------------------|-------|-------|
| female                 | 0.231     | 0.046              | 9.04  | 0.000 |
| pgssyear               | 0.033     | 0.006              | 10.22 | 0.000 |
| age                    | 0.027     | 0.005              | 6.70  | 0.000 |
| age2                   | -0.001    | 0.001              | -9.90 | 0.000 |
| N                      | 8588      |                    |       |       |
| log-lik                | 7877.3723 |                    |       |       |
| $R^2$ McKelvey-Zavoina | 0.092     |                    |       |       |
| Count- $R^2$           | 0.577     |                    |       |       |
| Adj. Count- $R^2$      | 0.024     |                    |       |       |

Significance level  $\alpha = 0.05$ .

1. What are assumptions of an ordered probit model?
2. Which Classical Linear Regression Model Assumption will plausibly be false for a linear probability model for this problem?
3. Interpret  $R^2$  statistics.
4. What is the interpretation of *pgssyear*'s estimate?
5. Interpret marginal effect for *female*.
6. The model was reestimated with dummy variables for voivoidship a respondent lives in. There are 16 voivoidships in Poland. The log-likelihood function value was equal to  $-7822.3119$ . Test the joint hypothesis that a voivoidships do not differ level of education.  
**Hint:**  $\chi^2_{13}(0.95) = 22.36$ ,  $\chi^2_{14}(0.95) = 23.68$ ,  $\chi^2_{15}(0.95) = 24.00$ .

RW : Calculate and interpret marginal effect of age for the alternative *higher education*.